Performance Prediction for Retrieval and Generative AI: A State-of-the-Art Review

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***Abstract*—Performance prediction anticipates instance-level success or failure before full evaluation or costly execution. This review connects four lines of work: Query Performance Prediction (QPP), Prompt Performance Prediction (PPP), RAG-oriented prediction, and human difficulty estimation in community question answering. It compares their evidence sources, prediction targets, evaluation practices, transfer limits, and supported decisions. Current work often treats retrieval effectiveness, generation quality, human difficulty, and routing cost as separate prediction problems; the review argues that they should be connected through a shared evidence–target–decision framework. Stack Exchange-style data provide a practical setting for testing where human difficulty traces and model outcomes align or diverge.**

***Index Terms*—Query Performance Prediction, Prompt Performance Prediction, information retrieval, large language models, retrieval-augmented generation, community question answering, prompt difficulty.**

I. Introduction

Modern information access systems often combine retrieval, generation, and evaluation in one pipeline. Average benchmark scores hide an instance-level reliability problem: a strong system can still fail on a particular query, prompt, context, or user request. A retriever may return weak evidence for one query; a generator may fail because the prompt is ambiguous, the evidence is insufficient, the output constraints are underspecified, or the evaluator rewards a different property. The reliability question is therefore not only how the system performs on average, but whether the current input is likely to succeed before the system produces a costly or trusted answer.

Performance prediction addresses this uncertainty. In information retrieval, Query Performance Prediction (QPP) estimates the expected effectiveness of a retrieval run for a given query, collection, retriever, ranker, and metric [1], [2]. In generative AI, Prompt Performance Prediction (PPP) is used here as a review label for the prompt-focused counterpart, covering both direct PPP work and studies that provide evidence for predicting prompt success.

This review treats performance prediction as a controlled-decision problem: evidence available at a given point in the pipeline is used to predict a stated target and choose an action under cost and robustness constraints. Three mismatches motivate this view. First, there is a *retrieval–generation disconnect:* QPP can estimate ranked-list effectiveness, but RAG systems need to know whether retrieved material will support a faithful generated answer [3], [4]. Second, there is a *human–model difficulty disconnect:* community question-answering traces reveal which questions are difficult for humans, but model difficulty is usually measured separately through correctness, faithfulness, uncertainty, abstention, or cost [5], [6], [7]. Third, there is a *prediction–decision disconnect:* QPP studies have long shown that score prediction, error analysis, and selective processing are not equivalent to downstream utility [8], [9].

The contribution is threefold.

1) a classification of performance-prediction work across QPP, PPP, RAG-oriented prediction, and human question difficulty;
2) a comparison of method families by evidence, target, timing, evaluation practice, robustness limit, and supported action;
3) benchmark implications for paired human–model studies on Stack Exchange-style data.

In both QPP and PPP, difficulty is not an intrinsic property of the input string. It is defined by the input, the system configuration, the evaluation target, and the action being considered.

The relevant actions include reformulating a query, switching retrievers, selecting a prompt variant, generating additional samples, invoking a judge, requesting clarification, routing to a stronger model, or abstaining. Stack Exchange-style threads provide authentic question instances with both community difficulty traces and prompt-like model-evaluation inputs.

II. Review Scope and Search Protocol

Following the definition above, the scope is bounded by one question: how can a system predict whether the current query or prompt will succeed in its pipeline? This boundary excludes a full survey of adjacent areas such as query expansion, learning to rank, prompt engineering, prompt tuning, instruction tuning, hallucination detection, model routing, human evaluation, and safety evaluation. Adjacent areas are included only when they directly support the review framework introduced in Section I.

Search Strategy. The corpus was assembled from seed papers in QPP, prompting, and RAG, followed by backward and forward snowballing through ACM Digital Library, ACL

TABLE I
CORPUS BLOCKS, INCLUSION CRITERIA, AND REVIEW ROLE.

Topic block	Main sources	Inclusion rule	Corpus role
Classical QPP	ACM Digital Library, Springer, citation links, seed monographs	Defines QPP concepts, method families, or evaluation practice	retrieval foundation
Neural and RAG-oriented retrieval	ACM Digital Library, ECIR/SIGIR proceedings, arXiv, citation links	Studies transfer, neural retrieval, RAG utility, or selective control	retrieval-generation link
PPP-relevant generation evidence	ACL Anthology, OpenReview, arXiv, publisher pages	Provides direct PPP evidence or measurements for prompt success, robustness, judging, or cost	generation prediction
CQA and Stack Exchange difficulty	ACL Anthology, ACM Digital Library, publisher pages, targeted web search, Stack Exchange documentation	Supports human difficulty measurement or human–model comparison	difficulty grounding
Adjacent control topics	citation links and targeted searches from selected papers	Defines evidence, targets, or actions for controller design	control evidence

Anthology, arXiv, Springer, publisher pages, and citation links. It was then extended with targeted searches for CQA question difficulty, Stack Overflow difficulty and answerability, Stack Exchange data fields, LLM answers to Stack Overflow, and human–model difficulty concordance.

Inclusion and Exclusion Criteria. References were included when they served one of five roles: defining a core concept, introducing a method family, exposing an evaluation or robustness limitation, supporting a control decision, or grounding human difficulty measurement. Work focused only on improving retrieval or generation without such a role was excluded. This selection keeps the review focused on performance prediction across retrieval, prompting, and RAG.

Corpus Grouping. The corpus is grouped by review role rather than recency alone. Table I summarizes the source blocks, inclusion criteria, and review role for each block. Each reference was also assigned one primary role for Figure 1, although several support more than one local claim.

Representative search terms included query performance prediction, query difficulty, neural IR QPP, RAG retrieval utility, query variant selection, prompt performance prediction, prompt robustness, LLM judge, self-consistency, CQA question difficulty, Stack Overflow difficulty, unanswered questions, LLM answers to Stack Overflow, and Stack Exchange schema. The corpus includes peer-reviewed papers, books or monographs, official documentation, and recent preprints or to-appear papers when they directly support this scope. Citations are used locally: when a paragraph names a method, dataset, result, or limitation, the adjacent reference supports that statement.

The organization follows prediction timing, evidence source, prediction target, and supported decision. Chronology remains relevant because QPP developed before PPP and because RAG changes the target of prediction, but the shared structure is clearer when methods are compared by role.

Figure 1 shows the resulting distribution by period and role.

For readability, the main technical abbreviations used in the tables are defined here. Retrieval terms are IDF (inverse document frequency), ICTF (inverse collection term frequency), SCQ (a simplified clarity-style query statistic), WIG (Weighted Information Gain), AP (Average Precision), nDCG (normalized Discounted Cumulative Gain), sMARE (a QPP error-analysis measure), BM25 (a standard lexical retrieval model), qrels (relevance judgments), and SEDE (Stack Exchange Data Explorer). Generative-AI terms are groundedness (support from provided evidence), faithfulness (consistency between an answer and that evidence), calibration (alignment between confidence and correctness), and LLM-as-judge (a language model used as an evaluator). Numeric values show the scale of selected source results.

III. QUERY PERFORMANCE PREDICTION

Building on Section I, QPP has developed through three broad stages: lexical uncertainty, ranked-list diagnostics, and selective control in neural and RAG pipelines. Under that system-dependent definition, the same query can be easy or difficult depending on the collection, representation, retriever, ranker, and metric [10], [11].

A. *Background: Query Difficulty and Vocabulary Mismatch*

The roots of QPP lie in the vocabulary problem. Furnas et al. showed that different people naturally choose different words for the same concept [12]. In information retrieval, this implies that a relevant document may exist even when it does not contain the words used in the query. Retrieval failure can therefore arise from language variation rather than from an impossible information need or an absence of relevant documents.

Classical QPP asks whether such failures can be anticipated. A short query with ambiguous terms may produce scattered results; a query dominated by common terms may retrieve many weakly related documents; and a query whose vocabulary does not match the collection may miss relevant material. The objective is not to retrieve documents directly, but to estimate whether the retrieval process is likely to succeed and whether an intervention is justified.

Carmel et al. make this system dependence explicit by treating query difficulty as a function of the query, collection, and relevance evidence [10]. Zendel et al. extend the distinction by separating information needs, query formulations, and systems [11]. This distinction matters whenever several formulations express the same need.

B. *Pre-Retrieval Predictors*

Pre-retrieval QPP predicts difficulty before the ranked list is produced. He and Ounis showed that query terms and collection statistics can provide evidence for retrieval-time risk before ranking [13]. Zhao et al. studied similarity and

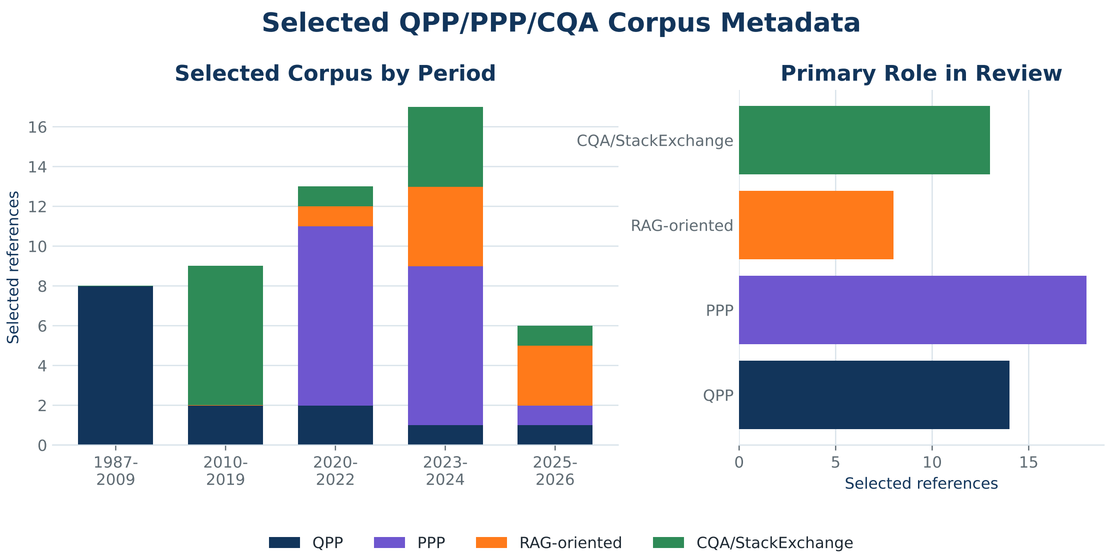

Fig. 1. **Distribution of selected references by publication period and corpus role.** Counts come directly from BibTeX metadata and role labels.

variability evidence, showing that term behavior across the collection can support early prediction [14].

Pre-retrieval QPP is useful when the system must act before ranking. It can guide early interventions such as reformulation, retriever choice, or clarification, but it cannot observe how the retriever actually responds to the query. A query can contain specific terms and still fail because the collection lacks the required documents, because dense retrieval maps it poorly, or because relevant evidence is present but not ranked sufficiently high.

Linguistic QPP adds interpretive evidence to early prediction. Mothe and Tanguy show that query difficulty can be related to linguistic properties such as polysemy and syntactic structure [15]. This evidence is useful when the decision is clarification, reformulation, or expansion; Table II summarizes the broader trade-offs.

C. *Post-Retrieval Predictors*

Post-retrieval QPP shifts attention from the query alone to the system's first response. The clarity score is the foundational method in this family [1]. It compares the language model of the top-ranked documents with the background collection model. When the top documents form a focused and distinctive model, the query is considered clearer; when they resemble the general collection, the query is less clear. The method operationalizes a simple intuition: coherent top results suggest that the retrieval system has captured the intended topic.

Weighted Information Gain (WIG) and related score-distribution predictors follow a similar principle [16]. When top scores are clearly separated from the background or from lower-ranked documents, the ranking may be more reliable; when the distribution is flat or unstable, the query may be difficult. These methods are practical because they reuse information already produced by retrieval and can be inserted naturally into multi-stage systems.

The cost of post-retrieval prediction is the requirement of an initial retrieval run. This is too late for deciding whether retrieval should be attempted, but early enough to guide second-stage processing such as reranking, feedback, or fallback handling. The main risk is score dependence. A BM25 score, a dense dot product, and a neural reranker score do not have the same interpretation; a predictor calibrated on one retrieval family may not transfer to another.

D. *Learning-Based and Neural Predictors*

Learning-based QPP replaces single hand-designed predictors with supervised combinations of features. Such models can integrate query statistics, linguistic properties, top-document coherence, score distributions, and retrieval-stage metadata. Yom-Tov et al. connect such prediction to practical applications such as missing content detection and distributed information retrieval [17]. The strength of learning-based prediction is flexibility; its weakness is dependence on judged data and susceptibility to overfitting to a collection, metric, or retrieval stack.

Neural retrieval changes the assumptions behind many classical predictors. Dense retrievers represent queries and documents as vectors, learned sparse retrievers alter lexical evidence, and neural rerankers introduce model-specific scores. The issue is not merely a new ranker: score variance, term specificity, and result coherence can change meaning under learned representations.

Selective query processing makes the control role explicit. The system applies an expensive intervention only when prediction suggests that it is likely to help, such as spending extra computation on expansion, reranking, routing, or query variants. Robustness matters because a predictor that

TABLE II
MAIN FAMILIES OF QPP METHODS AND THEIR PRACTICAL TRADE-OFFS.

Family	Timing	Main evidence	Typical purpose	Main limitation
Linguistic predictors	before retrieval	query length, syntax, morphology, ambiguity	explain formulation difficulty	language and query-type dependence
Statistical pre-retrieval predictors	before retrieval	IDF, ICTF, SCQ, collection statistics	cheap early warning	no view of actual ranking behavior
Clarity and coherence predictors	after first retrieval	language-model divergence, result focus	detect ambiguous or scattered results	requires a first retrieval run
Score-distribution predictors	after first retrieval	WIG, score magnitude, score variance	estimate ranking confidence	scores may not transfer across models
Learning-based predictors	before or after retrieval	combinations of query and result features	adapt several weak signals to a target metric	needs judged training data
Neural and dense predictors	before or after retrieval	embeddings, vector neighborhoods, model agreement	align QPP with semantic retrieval	calibration and transfer are difficult
Selective-processing predictors	at action time	any QPP signal tied to a policy	decide when to reformulate, rerank, or switch systems	value depends on action and cost model
RAG-oriented predictors	before generation or after retrieval	query variants, retrieved context, answer utility signals	select variants or predict generation benefit	requires downstream validation

correlates with effectiveness in one setup may still fail as a controller in another; Table XII keeps the neural-transfer and collection/ranker reference points together.

Table II consolidates the timing, evidence, purpose, and limitation trade-offs across the method families discussed above.

E. Evaluation Metrics and Benchmarks

QPP is commonly evaluated by comparing predicted query scores with observed query-level effectiveness. Correlation metrics remain useful diagnostics because they test whether predicted difficulty tracks retrieval outcomes, but they do not by themselves show whether a predictor improves a retrieval decision.

The deployment question is threshold-sensitive: a predictor can order queries well and still select poor cases for intervention. Faggioli et al. therefore propose sMARE to inspect prediction error behavior rather than relying on a single aggregate correlation [8]. Ganguly et al. show that QPP conclusions vary with metrics, collections, rank depth, and systems [19]. These results raise the reporting standard: a QPP study should report the setting, target metric, retrieval model, query set, and downstream decision it supports.

Common QPP evaluation criteria differ in what they measure and what they miss. Table III stays QPP-specific; Section VIII generalizes the decision-oriented evaluation design.

The operational test is the threshold at which the predictor triggers a defined intervention.

TABLE III
EVALUATION CRITERIA FOR QPP AND WHAT EACH CRITERION MISSES.

Evaluation criterion	What it measures	Why it is useful	What it misses
Pearson correlation	Linear association between predicted and true performance	Simple standard comparison	Non-linear decisions and threshold behavior
Spearman correlation	Rank agreement between predicted and true query order	Useful for ordering queries by difficulty	Does not measure action utility
Kendall correlation	Pairwise ordering agreement	Robust for comparing query orderings	Can hide large absolute errors
AP or nDCG target	Prediction of a ranked-list metric	Connects QPP to IR evaluation	May not match user or RAG utility
sMARE-style analysis	Error distribution and method behavior	Shows instability and variation	Less direct as a system objective
Selective utility	Gain from applying actions to selected queries	Tests practical usefulness	Depends on action cost and threshold
RAG downstream utility	Effect on generated answer quality	Measures final pipeline value	Expensive and generator-dependent

TABLE IV
WHY QPP BECOMES HARDER IN NEURAL AND RAG-ORIENTED RETRIEVAL.

Setting	Main retrieval signal	QPP difficulty	Consequence
Sparse retrieval	lexical matching and term statistics	signals are relatively interpretable	classical predictors remain informative baselines
Learned sparse retrieval	model-generated lexical representations	expansion changes query evidence	predictors must account for learned term behavior
Dense retrieval	vector similarity	scores are hard to calibrate across models	embedding and robustness signals are needed
Hybrid retrieval	sparse and dense fusion	components can fail differently	QPP can support retriever or fusion selection
Reranking	neural rescoring of candidates	candidate generation and reranking interact	prediction may need multi-stage evidence
RAG	retrieved context used by a generator	relevance and answer utility can diverge	QPP must target downstream utility

F. Selective Retrieval and Query Reformulation

Modern retrieval systems are heterogeneous: sparse lexical retrieval, dense retrieval, hybrid fusion, neural reranking, and RAG each expose different confidence signals and failure modes. Table IV keeps the setting-by-setting comparison, while the prose here focuses on the control consequence.

The control consequence is that a confidence signal is meaningful only for a particular retrieval family and downstream use case. Selective processing therefore ties each intervention to a predicted failure mode, so the predictor must be evaluated against the selected action.

G. *QPP for Retrieval-Augmented Generation*

RAG changes the QPP target from ranked-list effectiveness to downstream answer utility. Section VI-B discusses this retrieval–generation coupling in detail. The cited RAG work establishes the target shift, while Table XII keeps the empirical anchors separate [20], [3], [4], [21], [22], [23].

IV. Prompt Performance Prediction

This section uses PPP, defined in Section I, to group prompt-side evidence for predicting generation success. It distinguishes direct PPP methods from broader evidence sources: prompt design, calibration, prompt search, robustness, response sampling, factuality checking, LLM-as-judge evaluation, RAG evaluation, and cost-sensitive deployment. Under the same view, the prediction object is the prompt–system relation rather than the prompt string alone.

A. *Background: Prompting and Task Representation*

Few-shot prompting made prompt design central by showing that in-context examples can induce task behavior without parameter updates [25]. Liu et al. describe the broader paradigm as "pre-train, prompt, and predict," in which task formulation is moved into the input [26]. Pattern-Exploiting Training and LM-BFF further show that prompt patterns, verbalizers, and demonstrations shape how classification tasks are represented for language models [27], [28].

These results imply that the prompt is not a neutral wrapper around the task. It is an interface between the intended operation and the model's learned behavior. When this interface is poorly aligned, the model may fail despite having relevant knowledge; when it is well aligned, the model may perform a task from only a few examples. PPP begins at this interface: it asks whether the current formulation is likely to activate the desired behavior under the current model and evaluation target.

B. *Prompt Sensitivity and Calibration*

Prompt performance is often sensitive to details that appear minor. Zhao et al. show that few-shot prompts can induce label priors and that calibration can reduce these effects [29]. Lu et al. show that the order of examples can affect model behavior [30]. Sclar et al. show that formatting choices can produce large performance differences in their studied few-shot setting [31]. PromptRobust adds an adversarial perspective by evaluating robustness under prompt perturbations [32].

Automatic prompt search creates candidate prompts that must be ranked or rejected. AutoPrompt searches discrete trigger tokens [33]; RLPrompt optimizes prompts through reinforcement learning [34]; and Automatic Prompt Engineer uses LLMs to propose and evaluate instruction candidates [35]. Section VI-C returns to the broader variant-selection analogy.

VAP3 is a direct PPP example: it estimates before execution whether an LLM will generate a correct response for a prompt, and uses prompt variations to improve robustness to trivial modifications [24]. Robustness, judge, sampling, and factuality studies play a different role: they supply signals for prediction or control without necessarily proposing a PPP method.

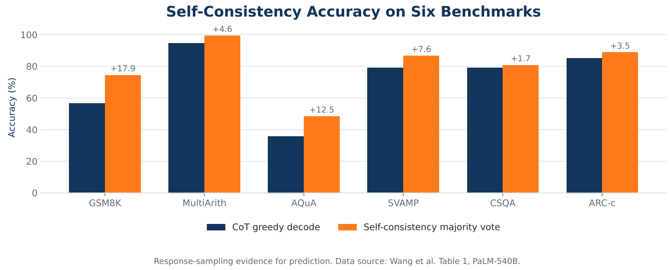

Fig. 2. **PaLM-540B accuracy for chain-of-thought greedy decoding and self-consistency majority vote on six reasoning benchmarks.**

The important distinction is between empirical prompt performance and robust prompt performance. A prompt can score well on a small development set because it exploits a template, label verbalizer, or benchmark artifact. Robustness tests should therefore perturb paraphrases, formatting, example order, models, and evaluation targets. Robust prompt evaluation asks whether performance remains reliable under the variation expected in the intended setting, not only whether one template works once.

C. *Reasoning and Response-Level Signals*

Some PPP evidence becomes available only after the model begins to answer. Chain-of-thought prompting asks the model to produce intermediate reasoning [36]. Zero-shot reasoning prompts show that short instructions can elicit reasoning behavior without demonstrations [37]. Self-consistency samples multiple reasoning paths and uses answer agreement as a stability signal [38]. Section VII generalizes this timing distinction across formulation, sampling, evaluation, and cost signals.

Figure 2 gives the empirical reference point from Wang et al.'s Table 1 [38]. Prediction validity still requires testing whether response variation predicts correctness, faithfulness, or abstention value.

Response-level evidence is risk evidence, not a correctness certificate. Agreement can reflect shared bias, fluent reasoning can be wrong, and one prompt can yield both correct and hallucinated samples. SelfCheckGPT addresses a related factuality problem by comparing sampled responses from a black-box model to detect inconsistent factual claims [39]. PPP should therefore use self-consistency, self-checking, and sample disagreement to trigger further checking.

D. *Automatic Evaluation and LLM Judges*

Open-ended generation requires evaluation beyond exact match, and this makes the evaluator part of the prediction problem. Judge signals may come from human, metric-based, classifier-based, or LLM-based evaluators. G-Eval uses GPT-4 with structured criteria and reports higher alignment with human judgments than several conventional metrics in its studied settings [40]. MT-Bench and Chatbot Arena study LLM-as-judge evaluation and expose both scalability advantages

TABLE V

SELECTED QUANTITATIVE VALUES FOR PPP EVIDENCE.

Source	Reported quantity	PPP relevance	Limitation for prediction use
Wang et al. [38]	Figure 2 reproduces six reported reasoning-benchmark accuracies.	Provides response-sampling evidence.	Agreement is model- and task-specific and can still be wrong.
Liu et al. [40]	G-Eval with GPT-4 reports Spearman correlation 0.514 with human judgment on summarization.	Supports judge-based PPP as a scalable but target-specific evaluation signal.	One evaluator setting; judge scores are surrogate targets.
Zheng et al. [41]	GPT-4 judges report over 80% agreement with human preferences; released resources include 3K expert votes and 30K preference conversations.	Shows why LLM-as-judge evidence can support prompt acceptance or escalation decisions.	Preference agreement does not establish correctness, faithfulness, code executability, or absence of judge bias.
Zhu et al. [32]	PromptRobust reports 4,788 adversarial prompts across 8 tasks and 13 datasets.	Makes robustness testing a measurable PPP input rather than an informal warning.	Benchmark scale does not quantify robustness for every deployment prompt.
Sclar et al. [31]	Formatting variants produce performance differences up to 76 accuracy points with LLaMA-2-13B in their few-shot setting.	Shows that plausible formatting variants can materially change prompt performance.	The reported maximum is model- and setting-specific.
Saad-Falcon et al. [42]	ARES evaluates RAG systems across 8 knowledge-intensive tasks while using a few hundred human annotations during evaluation.	Supports target decomposition for RAG-oriented prediction into context relevance, answer faithfulness, and answer relevance.	Primarily an evaluation framework; pre-generation use would require additional predictor design.

and bias risks [41]. ARES decomposes RAG evaluation into context relevance, answer faithfulness, and answer relevance [42]. These works show that PPP predicts a target-specific evaluation outcome; the target taxonomy is defined in Section IV-F.

Table V gives selected quantitative anchors for these PPP signals.

E. Prompt and Context Features

Prompt and context features turn the sensitivity results above into observable variables. Prompt text is the cheapest PPP evidence source because it is available before generation: task statement, output schema, delimiters, role instructions, refusal rules, evidence-use instructions, and possible contradictions. Such checks provide early diagnostic warnings but not complete behavior prediction.

Surface clarity still does not imply model alignment. PET and LM-BFF show that task performance can depend on the pattern and verbalizer used to map an input to labels [27], [28]. A human-readable prompt can map poorly to the model's learned distribution, whereas an automatically discovered prompt can perform well despite being unintuitive [33]. PPP therefore treats prompt text as evidence, not as a complete explanation.

Few-shot examples should be recorded as feature fields rather than folded into a generic prompt string. Coverage, similarity to the target input, diversity, label balance, order position, and format consistency expose the parts of the prompt that Section IV-B identifies as unstable.

Retrieved context is the RAG-specific counterpart. It can be inspected before generation through passage relevance, source quality, factual consistency, sufficiency, length, and placement. Long-context findings show that relevant information can be present but used poorly depending on its position [43]. Section VI-B discusses the resulting failure modes.

F. Prediction Targets

Prompt success is inherently multi-target. Correctness concerns task solution; helpfulness concerns user value; groundedness concerns support from provided evidence; factuality concerns the truth of generated claims; instruction following concerns constraints such as length, format, refusal, or citation style; safety concerns harmful or policy-violating behavior; and cost concerns resource use.

These targets can conflict. A chain-of-thought prompt can improve some reasoning tasks while increasing token cost [36]. Self-consistency can improve stability while multiplying generation calls [38]. A RAG prompt can include more evidence while making the relevant passage harder to use if it is placed poorly in a long context [43]. A prompt can also be optimized for an LLM judge without improving the intended human or task-specific target [41].

A PPP predictor must state whether it estimates correctness, preference, factuality, groundedness, safety, format validity, cost, or a decision-specific combination. Different deployments then weight those targets differently: factual systems emphasize groundedness, tool pipelines emphasize format validity, and production systems emphasize utility per unit cost.

G. Cost-Sensitive Routing

Cost-sensitive PPP connects evaluator signals to routing under an explicit cost model. A judge call, response sample, query rewrite, or stronger-model invocation is itself an action with resource and evaluator-bias costs. Section VIII-E discusses cost accounting.

V. HUMAN DIFFICULTY IN COMMUNITY QUESTION ANSWERING

This section turns to community question answering as evidence for human question difficulty. Section V-D later connects those traces to model outcomes through a benchmark protocol.

A. Question Difficulty in CQA

CQA research provides direct precedents for estimating question difficulty. Liu et al. model question difficulty through competition between askers, questions, and answerers, and show that question descriptions themselves contain difficulty evidence [44]. Wang et al. extend this line with a regularized model that combines interaction evidence with textual descriptions, which also addresses cold-start questions with few interactions [45]. QDEE jointly estimates question difficulty and user expertise, using the observation that users tend to ask and answer more difficult questions as their expertise evolves [46]. DiffQue reformulates the problem as relative difficulty estimation between pairs of questions, which is useful because annotators can often compare two questions more reliably than assign an absolute difficulty level [47].

Stack Overflow studies make the connection more concrete by operationalizing difficulty through manual labels, early question features, later response behavior, answerability, and latency [5], [48], [49], [50], [51]. Table VII lists the selected numeric reference points.

This connection also fits broader work on text-based difficulty estimation, which treats difficulty as a property inferred from language, task demands, and learner or user context rather than from text length alone [52]. The public dump and Stack Exchange Data Explorer schema expose the metadata required for this study [53]. Stack Exchange traces are treated as indirect evidence, not labels: answer status, latency, comments, votes, closure, and accepted-answer behavior must be interpreted with site, tag, traffic, moderation, and temporal controls.

B. Human–Model Difficulty Alignment

Treating a Stack Exchange question as a prompt creates a realistic human–model comparison under explicit controls. Table VII gives the Stack Overflow reference point motivating separate validation of model outputs rather than reliance on surface helpfulness.

The human–model relation should also not be assumed to be monotonic. Difficulty-concordance work motivates reporting concordance, discordance, avoidance, and prompt stability rather than assuming one difficulty ordering [7].

The main Stack Exchange signals link data fields and judgments to the kind of difficulty they can support.

The benchmark therefore keeps platform evidence, difficulty labels, and validated model outcomes separate.

C. Difficulty Families

Prompt-like Stack Exchange difficulty can be classified into five multi-label families. A single thread can be difficult for several reasons at once, so the families connect human traces to model risks and validation needs without forcing one exclusive label.

These families provide the bridge from CQA traces to the benchmark protocol below.

D. Implications for a Stack Exchange Benchmark

CQA and Stack Overflow literature suggest using the full Stack Exchange thread as the unit of analysis. The initial post supplies the prompt-like formulation; answers, comments, edits, votes, timestamps, and closure history provide human-difficulty evidence and platform controls. Table IX preserves this timing distinction so that early predictors can use question text and tags, while later predictors can add retrieval evidence, response samples, judges, and cost traces. Model difficulty should be measured through the outcome and evaluation fields in Table IX, rather than inferred from community behavior alone.

Figure 3 summarizes the paired record structure used to analyze concordance and discordance between human and model difficulty without collapsing those label spaces. Trace threats from Table VI and provenance fields from Table IX remain controls rather than difficulty labels.

TABLE VI
STACK EXCHANGE SIGNALS FOR CONNECTING HUMAN QUESTION DIFFICULTY TO MODEL PROMPT DIFFICULTY.

Signal	Human interpretation	Model analogue	Main threat
No answer or no accepted answer	unresolved or unattractive to answerers	model fails, abstains, or gives unsupported answer	asker may not accept a correct answer; visibility varies
Time to first or accepted answer	community effort or expert scarcity	additional samples, tools, or retrieval needed before correctness	latency depends on traffic and time of posting
Answer count and disagreement	several possible solutions or contested answer space	response instability across generations or models	many answers can indicate popularity, not difficulty
Comments before answer	need for clarification, missing context, or debugging negotiation	model asks follow-up questions or makes assumptions	comments can include partial answers
Closure, duplicate, or migration history	unsuitable, too broad, unclear, off-topic, or already solved elsewhere	prompt violates task boundary or needs reformulation	closure captures community norms, not only difficulty
Tags, topic, and site	domain, expertise pool, and expected answer style	model domain competence and retrieval need	tag popularity confounds difficulty
Question text, code, links, and formatting	intrinsic formulation and reproducibility evidence	prompt completeness, executable context, and instruction clarity	surface clarity may not imply answerability
Manual difficulty label	explicit human difficulty judgment	target for human–model concordance analysis	annotation requires a stable rubric and agreement checks
Model correctness, faithfulness, and reproducibility	not a human signal	primary model-difficulty outcome	accepted Stack Exchange answer may be incomplete or outdated

TABLE VII
SELECTED EMPIRICAL VALUES FOR STACK EXCHANGE-BASED PROMPT DIFFICULTY.

Source	Reported value	Use for human–model difficulty analysis
SOQDE [5]	936 Stack Overflow questions were manually labeled as basic, intermediate, or advanced; Random Forest reached 67.6% mean accuracy with pre-hoc features and 75.2% with post-hoc features.	Provides a direct baseline for Stack Overflow human difficulty labels and shows the value of separating early and late evidence.
Raida et al. [48]	1,245 manually labeled Stack Overflow questions were classified using semantic and contextual features, with a distinction between a priori, pre-hoc, and post-hoc feature sets.	Supports a corpus design where immediate prompt text, asker/profile context, and later community signals are kept separate.
Mondal et al. [50]	The study analyzes 4.8 million Stack Overflow questions and reports models that predict unanswered questions during submission with about 79% maximum accuracy.	Shows that answerability can be predicted early, but unanswered status still requires qualitative interpretation.
Kabir et al. [6]	The study evaluates ChatGPT answers to 517 Stack Overflow questions and reports 52% incorrect answers, 77% verbose answers, and user preference despite overlooked misinformation in about 39% of cases.	Shows why model difficulty should be judged by correctness and verification, not only by fluency or user preference.
Zhou et al. [7]	The study compares correctness, avoidance, and prompt stability across human-calibrated difficulty levels and several model families.	Provides a methodological template for difficulty concordance, but not direct Stack Exchange evidence.

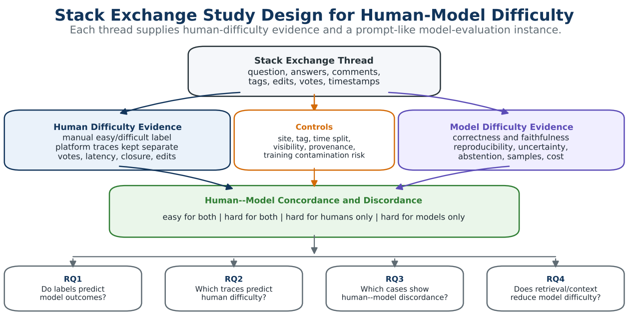

Fig. 3. **Human–model comparison record structure.** The diagram separates human evidence, model evidence, controls, and research questions.

TABLE VIII
OPERATIONAL DIFFICULTY FAMILIES FOR STACK EXCHANGE PROMPT-LIKE INSTANCES.

Difficulty family	Human trace	Model risk	Control or validation need
Formulation difficulty	unclear goal, missing context, clarification comments, incomplete example	wrong assumptions, weak follow-up behavior, hallucinated missing details	manual rubric for ambiguity and missing information
Evidence and context difficulty	links, edits, long thread context, answer-bearing comments, accepted-answer evidence	ignores relevant context or uses unsupported facts	context-sufficiency and faithfulness checks
Expertise and reasoning difficulty	advanced tags, expert scarcity, long answer latency, multi-step explanations	wrong reasoning, poor domain routing, unstable samples	site/tag stratification and expert validation
Execution and recency difficulty	code, environment details, API version, outdated answers, external documentation	non-reproducible answer, obsolete API, false tool or package claim	reproducibility and version checks
Evaluation and safety difficulty	disputed answers, closure, comments correcting answers, unsafe or policy-sensitive content	plausible but wrong answer, unsafe answer, bad refusal or overconfidence	human validation and judge-audit protocol

TABLE IX
MINIMUM PROTOCOL FOR A STACK EXCHANGE HUMAN–MODEL DIFFICULTY BENCHMARK.

Record block	Fields to preserve	Prediction role	Validity control
Thread provenance	site, post id, creation date, edit history, license, dump or API version	supports time splits and reproducibility	prevent leakage and undocumented filtering
Question formulation	title, body, code, links, tags, accepted formatting, closure or duplicate state	early human and model difficulty evidence	separate intrinsic content from later community behavior
Human traces	Table VI signal fields, timestamps, and allowed site or user metadata	late platform evidence for difficulty and answerability	interpret through Table VI threats and manual labels
Manual label	easy/difficult label, rubric, annotator agreement, uncertainty note	target for human difficulty modeling	require stable annotation rules and adjudication
Retrieval record	corpus, retriever, chunking, query variant, top passages, rank scores, context order	QPP and RAG-context prediction evidence	preserve settings that affect downstream generation
Prompt record	prompt template, examples, system instruction, output schema, retrieved context, model version	PPP evidence and reproducibility	report all instructions and context placement
Model outcome	answer, refusal, uncertainty, samples, tool calls, executable result when applicable	model difficulty target	validate correctness separately from fluency
Evaluation and cost	human judgment, judge prompt, faithfulness score, and resource logs	downstream value and cost-aware analysis	audit judge bias and report prediction cost

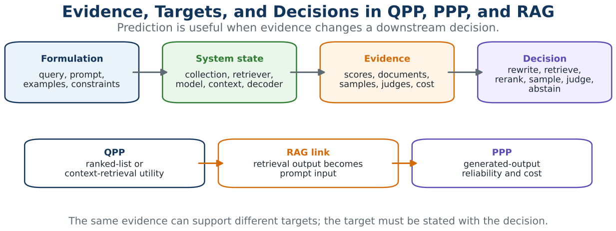

Fig. 4. **Comparison variables for QPP, PPP, and RAG.** The figure summarizes formulation, system state, evidence, target, and decision points across the three settings.

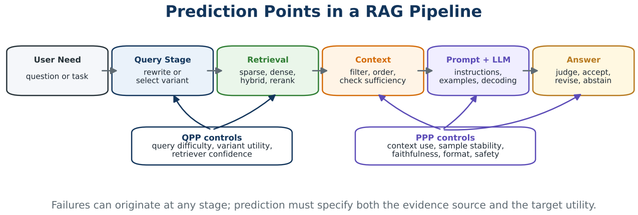

Fig. 5. **RAG intervention points.** The figure locates where predictors can observe evidence or change control decisions across a RAG run.

VI. Comparison: Evidence, Targets, and RAG

The earlier sections introduced QPP, PPP, RAG, and Stack Exchange difficulty separately. This section formalizes the shared comparison. Four questions recur: what formulation and system state define the problem, when evidence is available, what target is being predicted, and what action can change as a result. These questions can be compressed into a three-part comparison: evidence, target, and decision. A prediction setting can be described by evidence available at time t, system state, target, predictor, decision policy, cost model, and downstream value:

$$\boldsymbol{\mathcal{E}_t}, \boldsymbol{S}, \boldsymbol{Y}, \boldsymbol{f_\theta}, \boldsymbol{\pi}, \boldsymbol{C}, \boldsymbol{U}.$$

Here $\boldsymbol{\mathcal{E}_t}$ is the evidence observable at time t, such as query terms, Stack Exchange traces, retrieved passages, prompt text, response samples, judge scores, or cost logs. $\boldsymbol{S}$ is the system state, including the collection, retriever, ranker, generator, decoder, evaluator, and context-construction policy. $\boldsymbol{Y}$ is the target to be predicted, such as ranked-list effectiveness, answer correctness, groundedness, faithfulness, uncertainty, abstention risk, or latency. A predictor estimates $\widehat{\boldsymbol{Y}} = \boldsymbol{f_\theta}(\boldsymbol{\mathcal{E}_t}, \boldsymbol{S})$. A decision policy selects an action $\boldsymbol{a} = \boldsymbol{\pi}(\widehat{\boldsymbol{Y}}, \boldsymbol{C})$ under cost model $\boldsymbol{C}$. The final criterion $\boldsymbol{U}(\boldsymbol{a}, \boldsymbol{Y}, \boldsymbol{C})$ records downstream value: the prediction is useful only if the action it selects improves quality, safety, or cost relative to a baseline policy.

The notation enforces three requirements. A score is meaningful only with its target $\boldsymbol{Y}$. A predictor is portable only when the system state $\boldsymbol{S}$ is specified. A correlation is useful for deployment only when it is tied to the decision policy $\boldsymbol{\pi}$ and cost model $\boldsymbol{C}$. The next subsections use these requirements to compare RAG coupling and variant selection.

With this notation in place, the remaining comparison focuses on the RAG-specific coupling between retrieval evidence, prompt context, generation, and evaluation.

A. *RAG Pipeline Coupling*

In RAG, retrieval output becomes prompt input. This coupling makes the prediction target depend on where evidence is observed and where an action can still change the run.

Two RAG-specific risks follow. First, evidence may be present but poorly placed or poorly instructed for the generator [43]. Second, query variants, rerankers, generator calls, and judges can multiply cost.

B. *Error Propagation*

The RAG pipeline is attractive because each component appears interpretable, but failures propagate across components. If the query is underspecified, the retriever may return passages that are topically related but not answer-supporting. If retrieval is strong but context construction is poor, relevant evidence may be buried among irrelevant passages or placed where the model uses it less effectively. If context is sufficient but the prompt is underspecified, the model may answer from prior knowledge or ignore citation constraints. If the answer is fluent but unsupported, an evaluator that checks only answer relevance may accept it.

Failures are compositional: QPP can flag missing or weak evidence, but prompt and answer-level checks are still needed to detect unsupported generation.

C. *Query and Prompt Variant Selection*

Query rewriting and prompt search create structurally similar selection problems. A query rewriter can generate several retrieval queries from the same information need [23]. A prompt optimizer can generate several prompt templates or instructions for the same task [33], [34], [35]. In both cases, executing or evaluating all variants may be costly, while selecting one variant without prediction can be brittle.

The shared question is not which variant looks best to a human reader, but which variant is likely to improve the selected target under the selected system. For QPP, the target may be ranked-list effectiveness, recall of answer-bearing passages, or downstream answer utility. For PPP, the target may be correctness, faithfulness, format validity, safety, or cost. Variant selection therefore gives the two fields a common problem: both require predicting performance among semantically related formulations rather than comparing unrelated inputs.

VII. Cross-Cutting Prediction Signals and Methods

Table X consolidates the signal sources introduced in Sections III–VI. It organizes them by decision timing: early signals are cheaper and more actionable, while later signals are richer but more system-specific and costly.

TABLE X
SHARED SIGNAL SOURCES FOR QPP, PPP, AND RAG-ORIENTED PREDICTION.

Signal source	What it observes	Predictive role	Example action	Main limitation
Query text	formulation evidence	early retrieval-risk estimate	reformulate or route	no ranking evidence
Community traces	CQA difficulty evidence	human difficulty and answerability estimate	stratify or select benchmark cases	platform controls required
Retrieval response	ranked-list evidence	retrieval confidence or drift estimate	rerank or retrieve more	retriever-specific calibration
Prompt text	instruction and schema evidence	prompt-risk estimate	rewrite or request missing information	model interaction unseen
Context signals	support and placement evidence	answer-support estimate	trim, reorder, or abstain	relevance is not faithfulness
Response samples	generation-stability evidence	uncertainty or instability estimate	sample again or switch model	agreement can still be wrong
Evaluator signals	target-judgment evidence	acceptance or revision estimate	accept, revise, or escalate	evaluator bias
Robustness tests	variation evidence	transfer-stability estimate	select stable variant	added evaluation cost
Cost signals	resource-use evidence	utility-per-cost estimate	stop early or allocate budget	requires value model

Section VIII turns this consolidated signal view into benchmark baselines and reporting requirements.

VIII. EVALUATION PRACTICES AND BENCHMARK IMPLICATIONS

The evaluation protocol follows directly from the framework in Section VI and the signal taxonomy in Section VII. A predictor should be assessed at three levels: whether it estimates the stated target accurately, whether its scores remain calibrated near decision thresholds, and whether the resulting policy improves quality, safety, or cost relative to baselines.

A. *Decision-Oriented Evaluation*

Section III-E's QPP warning extends to PPP and RAG: predictors should be tested against controlled actions, not only target-score correlation [8], [19].

B. *Baselines, Metrics, and Ablation Studies*

Table XI lists the baseline families, evidence sources, metrics, and decision tests needed to separate prediction quality from downstream value and cost.

Because the signal sources are heterogeneous, ablations should remove one signal family at a time and compare learned policies with fixed-cost or majority alternatives. Error analysis should reuse the human–model difficulty quadrants defined in Section V-D.

TABLE XI
EVALUATION DESIGN FOR DECISION-ORIENTED PERFORMANCE PREDICTION.

Baseline or model	Evidence used	Target examples	Metrics	Decision test
Random or majority baseline	none or class prior	human difficulty, model correctness, abstention	accuracy, AUROC, AUPRC	minimum sanity check
Human-trace baseline	tags, answer status, answer latency, comments, votes, closure	human easy/difficult label	AUROC, F1, calibration error	stratified sampling or annotation triage
QPP-only baseline	query text, collection statistics, top documents, rank scores	retrieval effectiveness or answer-bearing passage recall	Pearson, Spearman, Kendall, nDCG target, calibration	retrieve more, rerank, or rewrite
PPP-only baseline	prompt text, examples, output schema, model metadata, response samples	model correctness, format validity, uncertainty, refusal	AUROC, Brier score, ECE, selective risk	rewrite prompt, sample again, route model, or abstain
RAG-context baseline	retrieved context, source quality, sufficiency, placement, conflicts	groundedness, faithfulness, answer utility	faithfulness score, judge agreement, error type	reorder context, filter passages, retrieve more
Joint decision model	human traces, QPP signals, PPP signals, context evidence, samples, cost	human–model concordance and final utility	concordance rate, discordance analysis, utility-cost frontier	choose the least costly action that preserves quality
Oracle or full-context upper bound	full thread, all retrieved evidence, final answer, human or expert judgment	maximum available-information target	upper-bound score and latency/cost	quantify the gap closed by cheaper predictors

C. *Empirical Reference Points*

Table XII is the single location for study-specific empirical anchors in this review. It reports the scale of neural-transfer, RAG-evaluation, efficiency, and robustness effects that earlier sections discuss conceptually.

D. *Robustness Analysis*

QPP requires transfer across retrieval settings [19], [9]; PPP requires transfer across prompt variations and model settings [30], [31], [32]; RAG requires cross-pipeline tests because query variants, context placement, generation, and judging interact.

E. *Cost and Latency Analysis*

A predictor that correlates with effectiveness but costs more than the action it controls may be impractical. A cheap predictor that routes only uncertain cases to stronger processing can improve utility per cost, but reported costs should match the accounting fields in Table XIII.

In RAG and PPP, sampling, LLM-as-judge evaluation, and query rewriting can all improve evidence while multiplying computation or judge dependence [38], [40], [41], [23], [22].

TABLE XII
SELECTED EMPIRICAL REFERENCE POINTS FOR QPP/RAG PREDICTION.

Reference point	Reported basis	Why it matters for the comparison
Neural retrieval: up to 10%	Faggioli et al. report a drop of as much as 10% for QPP on neural models in semantic settings [18].	Classical evidence can lose calibration when retrieval representation changes.
RAG evaluation: $+0.168$ to $+0.494$	Salemi and Zamani report Kendall's τ improvement range for eRAG [3].	Retrieval prediction needs targets linked to generated-answer utility.
RAG efficiency: up to 50x	Salemi and Zamani report up to 50 times less GPU memory compared with end-to-end evaluation [3].	Performance prediction can reduce evaluation cost when it replaces expensive full-pipeline checks.
Robustness: collection $>$ ranker	Chifu et al. report that collection effects dominate ranker effects in their robustness analysis [9].	Prediction claims need collection and ranker context rather than one universal score.

The most effective predictor is therefore the one that improves the final decision under explicit quality and cost constraints.

IX. Validity Limits and Reporting Practices

Table XIII consolidates the reporting requirements implied by the preceding sections.

The table is the reporting instrument; the remaining caveats concern the scope of this review rather than additional checklist fields.

A. Limitations

This state-of-the-art review has three broader limits. First, the bibliography is selective and organized around performance prediction rather than exhaustive coverage of every adjacent field. Second, the quantitative values cited above come from different datasets, metrics, models, and evaluation settings, so they should be read as reference points rather than directly comparable effects. Third, model behavior, Stack Exchange communities, APIs, documentation, and accepted answers can drift over time, so benchmark conclusions need temporal qualification.

X. State-of-the-Art Verdict and Open Problems

The state of the art is uneven. Table XIV gives the verdict by literature family, separating current support from the main open problem.

The strongest open direction is a shared benchmark that evaluates predictors as controllers. For the Stack Exchange setting, that means preserving the record schema in Table IX, applying the reporting checklist in Table XIII, and testing transfer across the dimensions most likely to vary in deployment. The benchmark should prioritize target alignment, robustness under expected variation, and cost-aware value.

XI. Conclusion

Performance prediction should be judged by the decisions it improves. QPP contributes mature tools for estimating retrieval risk, while PPP and RAG require broader targets: generation reliability, context sufficiency, evaluator behavior, and cost-aware routing. Stack Exchange-style benchmarks can test this broader view because they expose human difficulty traces and model outcomes for the same authentic questions. The central challenge is not to produce a universal difficulty score, but to learn when a particular signal supports a particular action in a particular system.

TABLE XIII
REPORTING CHECKLIST FOR JOINT QPP/PPP STUDIES.

Item	What to report	Reason
Problem and target	Exact target, metric, and target family; use Section IV-F for PPP target types.	Different targets are not interchangeable.
Human difficulty evidence	For Stack Exchange/CQA traces and controls, use Tables VI and IX.	Human difficulty cannot be inferred from one platform variable alone.
Query or prompt object	Full query, variants, prompt template, examples, delimiters, output schema, and context instructions.	Needed to reproduce the sensitivity factors tested in Section IV-B.
Data provenance and time split	Report source, license, filters, time split; for Stack Exchange, use Table IX.	Public Q&A data can be affected by model-training contamination and site-specific norms.
System state	Collection, retriever, ranker, generator, model version, decoder, context window, and evaluator.	Results depend on this configuration.
Context construction	Retriever, corpus, chunking, passage count, ranking, ordering, and source quality for RAG.	Needed for RAG reproducibility.
Evaluation protocol	Metric, judge, judge prompt, human validation, qrels, faithfulness checker, or safety classifier.	Evaluator design defines the prediction target.
Robustness tests	Cross-collection, cross-ranker, cross-model, query variants, prompt paraphrases, formatting, attacks, and example order.	A single formulation is weak evidence for stability.
Cost accounting	Tokens, latency, samples, retriever calls, reranker calls, judge calls, and human review.	Prediction is useful only relative to its cost and decision.
Downstream value	Whether the predictor improves rewriting, retrieval selection, prompt selection, sampling, judging, abstention, or utility per cost.	Report the controlled action, not only the score.

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TABLE XIV
STATE-OF-THE-ART SYNTHESIS BY LITERATURE FAMILY.

Literature family	Representative sources	Current state	Main open problem
Classical QPP	Cronen-Townsend et al.; Carmel and Yom-Tov; He and Ounis; Zhao et al.; Zhou and Croft [1], [2], [13], [14], [16]	Mature pre- and post-retrieval predictors.	Transfer across collections, systems, query types, and metrics.
Neural and selective QPP	Zendel et al.; Faggioli et al.; Ganguly et al.; Chifu et al. [11], [18], [19], [9]	Transfer tests and selective-control studies.	Calibration under dense, hybrid, and learned retrieval.
RAG-oriented prediction	Lewis et al.; Salemi and Zamani; Tian et al.; Huly et al.; Arabzadeh et al.; Ma et al. [20], [3], [4], [21], [22], [23]	Retrieval evaluated as generator input.	Relevance, sufficiency, faithfulness, and utility diverge.
Prompt-side prediction and robustness	Brown et al.; Zhao et al.; Lu et al.; Sclar et al.; Arabzadeh and Bagheri [25], [29], [30], [31], [24]	Prompt variation as predictive evidence.	Standardized targets, benchmarks, and reporting.
Response, factuality, and judge evidence	Wei et al.; Kojima et al.; Wang et al.; Manakul et al.; Liu et al.; Zheng et al.; Saad-Falcon et al. [36], [37], [38], [39], [40], [41], [42]	Rich late-stage generation evidence.	Cost and evaluator dependence.
Human and CQA difficulty	Liu et al.; Wang et al.; Sun et al.; Thukral et al.; Hassan et al.; Raida et al.; Kabir et al.; Zhou et al. [44], [45], [46], [47], [5], [48], [6], [7]	Platform traces enable concordance studies.	Trace controls under site/time drift and contamination.

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